Price Path Embeddings in a Hang Seng Index Portfolio

Samson LAM

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Price path embeddings

- Returns over the past year are known to predict returns over the following months (Jegadeesh and Titman, 1993). Yet a year of daily returns and technical indicators holds far more inputs than a regression on noisy monthly returns can use.
- Gu et al. (2020) trace the gains of trees and neural networks over linear methods for US stock returns to nonlinear interactions, and find momentum among the dominant signals.
- Fischer and Krauss (2018) train a long short-term memory (LSTM) network on all S&P 500 return sequences to predict whether a stock beats the cross section the next day.
- The variational autoencoder (VAE) of Kingma and Welling (2014) embeds such an input in a latent space of low dimension whose coordinates can serve as forecasting variables.
- Principal component analysis (PCA) provides its linear counterpart.

Research question

Question

1. Do embeddings of the past year of returns and technical indicators of Hang Seng Index (HSI) stocks carry information about their next-month returns?
2. Does a nonlinear embedding improve on a linear one?
3. What do the embeddings deliver in a portfolio?

Methodology

1. The daily return path over the past year and technical indicators are embedded in two dimensions, linearly by PCA and nonlinearly by LSTM-VAE.
2. A pooled regression maps the embedding to a forecast of next-month returns.
3. The forecast enters the Black-Litterman model to construct the weights, and derivatives enhance yield through delta-hedged short calls and a conditional put spread collar.

Research question

Finding

We examine a USD 100 million long-only portfolio of HSI stocks with listed HSI derivatives, benchmarked against the HSI TR.

1. The embeddings carry significant information, with a rank information coefficient (IC) of about 0.07 and a t-statistic near 2.5.
2. The VAE and PCA perform similarly, because the standardized return path behaves like a random walk that leaves little structure beyond its principal components.
3. The PCA portfolio with derivatives strategies returns 6.28% a year against 3.09% for the HSI total return (HSI TR), an active return of 3.05% with an information ratio of 0.46.
4. The forecast is significant, but its transfer into active return is not, and neither are the derivatives strategies.

Monthly cycle of the strategy

- The portfolio is rebalanced at the close of the first trading day of each month from September 2021 to August 2026, the dates $\tau_1, \ldots, \tau_{60}$.

1. Embedding — At each rebalance, each stock's past year is embedded in low dimensions.
2. Stock selection — A pooled regression forecasts next-month returns from the embedding.
3. Portfolio construction — The Black-Litterman model turns the forecasts into weights.
4. Yield enhancement — Index derivatives are added to the resulting stock portfolio.

Price path representation

Both embeddings are fitted to the cumulative path of the standardized returns in the window,

$$c_{i,t,j} = \frac{1}{\sqrt{L}} \sum_{g=1}^{j} \frac{r_{i,t-L+g}}{s_{i,t}}, \qquad j = 1, \ldots, L.$$

- The input of stock i on day t is the window $X_{i,t} \in \mathbb{R}^{L \times 16}$ of its $L = 240$ most recent trading days, the sequence length of Fischer and Krauss (2018).
- Each day holds the 16 daily features of Wu et al. (2026), the return, open, high and low prices, volume and eleven technical indicators, made comparable across stocks.
- The path $\boldsymbol{c}_{i,t} = (c_{i,t,1}, \ldots, c_{i,t,L})$ records when the gains and losses occurred, and its end value $c_{i,t,L}$ is the return over the window scaled by its volatility.

Dimension reduction by PCA and the VAE

- PCA is linear and takes the return path alone.
- The VAE is nonlinear and takes the whole window, so it can find a low-dimensional curved surface on which the paths lie and draw on the technical indicators.
- The linear embedding $\boldsymbol{m}_{i,t} \in \mathbb{R}^2$ consists of the scores of $\boldsymbol{c}_{i,t}$ on its first two principal components, which carry 89.1% to 89.4% of the path variance at every refit.
- Both embeddings are refitted annually on the first trading day of each September from 2021 to 2025, on the windows before the refit.

Dimension reduction by PCA and the VAE

The VAE minimizes the negative evidence lower bound for a Gaussian likelihood of variance one half,

$$\mathcal{L}_{i,t} = \sum_{j=1}^{L} \left(c_{i,t,j} - \hat{c}_{i,t,j}\right)^2 + \mathrm{KL}\left(N(\boldsymbol{\mu}_{i,t}, \operatorname{diag} \boldsymbol{\sigma}_{i,t}^2) \,\|\, N(\mathbf{0}, I)\right).$$

- An LSTM encodes $X_{i,t}$ into the mean $\boldsymbol{\mu}_{i,t}$ and the log variance $\ln \boldsymbol{\sigma}_{i,t}^2$ of a Gaussian posterior over a latent $\boldsymbol{z} \in \mathbb{R}^d$.
- A second LSTM decodes a draw $\boldsymbol{z} = \boldsymbol{\mu}_{i,t} + \boldsymbol{\sigma}_{i,t} \odot \varepsilon$, with $\varepsilon \sim N(\mathbf{0}, I)$, into an estimate $\hat{\boldsymbol{c}}_{i,t}$ of the path.
- KL is the Kullback-Leibler divergence, and the VAE embedding is the posterior mean $\boldsymbol{m}_{i,t} = \boldsymbol{\mu}_{i,t}$, with $d = 2$ or $d = 3$.
- All stocks share a single network, as in the pooled design of Fischer and Krauss (2018).

Return prediction

The target is the excess return over the next month relative to the cross section, and a pooled regression on the embedding gives the forecast,

$$y_{i,t} = \sum_{j=1}^{21} \left(r_{i,t+j} - r^f_{t+j} \right), \qquad u_{i,t} = y_{i,t} - \bar{y}_t,$$

$$u_{i,t} = a + \boldsymbol{b}^\top \boldsymbol{m}_{i,t} + \nu_{i,t}, \qquad \hat{u}_{i,k} = \hat{a} + \hat{\boldsymbol{b}}^\top \boldsymbol{m}_{i,\tau_k - 1}.$$

- r^f_t is the daily return of cash at the 1-month Hong Kong Interbank Offered Rate fixing.
- $\bar{y}_t$ is the mean of $y_{i,t}$ over the HSI members on day t, following the cross-sectional target of Fischer and Krauss (2018).
- $\nu_{i,t}$ is the residual, and a and $\boldsymbol{b}$ are estimated at each refit by least squares pooled over the member days whose target ends before the refit.
- The forecast ranks the stocks, and its ranking power is measured by the rank IC and by the t-statistic of its mean over the 5 years.

Portfolio construction

$$\Pi_{i,k} = 21\,\delta\,\mathrm{Cov}\left(r_{i,t},\, r_{\mathrm{HSI},t}\right), \qquad \boldsymbol{Q}_k = \boldsymbol{\Pi}_k + \hat{\boldsymbol{u}}_k - \bar{u}_k \mathbf{1}, \qquad \Omega_k = \mathrm{diag}(\zeta \Sigma_k)$$

- The universe $\mathcal{U}_k$ at τ_k is the set of p_k HSI constituents with a year of complete returns and an embedding, from 57 to 93 stocks over the backtest.
- The prior is the equilibrium return implied by the HSI as the market portfolio, with $\delta = 2.5$ the risk aversion of He and Litterman (1999) and $r_{\mathrm{HSI},t}$ the daily return of the HSI.
- In each view, a stock beats its prior by its forecast relative to the universe mean $\bar{u}_k$.
- The uncertainty of the views is proportional to the prior covariance $\zeta \Sigma_k$ (He and Litterman, 1999). Σ_k is the 1-month covariance matrix, estimated from the 252 daily returns before τ_k with the noise eigenvalues of the correlation matrix cleaned.

Portfolio construction

With the identity matrix as the pick matrix, the posterior mean is

$$\boldsymbol{\mu}_k^{\mathrm{BL}} = \boldsymbol{\Pi}_k + \Sigma_k \left(\Sigma_k + \operatorname{diag} \Sigma_k\right)^{-1} \left(\hat{\boldsymbol{u}}_k - \bar{u}_k \mathbf{1}\right).$$

Each stock portfolio maximizes the mean-variance objective under a long-only constraint and the single-name cap of 3% in the investment proposal,

$$\boldsymbol{w}_k = \arg\max_{\boldsymbol{w}} \; \boldsymbol{w}^\top \boldsymbol{\mu}_k - \frac{\delta}{2} \boldsymbol{w}^\top \Sigma_k \boldsymbol{w} \quad \text{subject to} \quad \mathbf{1}^\top \boldsymbol{w} = 1, \;\; 0 \leq w_i \leq 0.03.$$

All proposed portfolios differ only in the expected returns $\boldsymbol{\mu}_k$.

Stock portfolios

Portfolio	Expected returns $\boldsymbol{\mu}_k$	Role
EW	none, $w_i = 1/p_k$	naive benchmark
M1	$\boldsymbol{\Pi}_k$	portfolio construction without stock selection
M2-PCA	$\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the PCA forecast	stock selection with the linear embedding
M2-VAE	$\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the VAE forecast, $d = 2$	stock selection with the nonlinear embedding
M3	$\boldsymbol{\mu}_k^{\mathrm{BL}}$ with the VAE forecast, $d = 3$	stock selection with a larger nonlinear embedding

Stock portfolios. All except EW solve the mean-variance objective with the stated expected returns.

- M1 against the HSI TR measures portfolio construction, and M2 against M1 measures stock selection.
- Every portfolio is rebalanced monthly and pays 15 basis points one way on turnover.

Derivatives strategies

The derivatives are written on the HSI, and each position holds

$$n = \left\lfloor \frac{h\,\beta_k\,E}{50\,S} \right\rfloor \text{ contracts.}$$

- E is the value of the stock portfolio and β_k its ex-ante beta to the HSI over the 252 days before τ_k.
- S is the HSI close, 50 the contract multiplier in HKD and h the coverage.
- We set the base case to $h = 0.5$, which covers 49% of the stock portfolio on average.
- Options are priced at the HSI Volatility Index (VHSI) level v_t by the Black-Scholes model with a continuous dividend yield, and every trade pays transaction costs.

Derivatives strategies

The yield strategy P1 targets the volatility risk premium. At a constant volatility, its hedged position gains, in HKD,

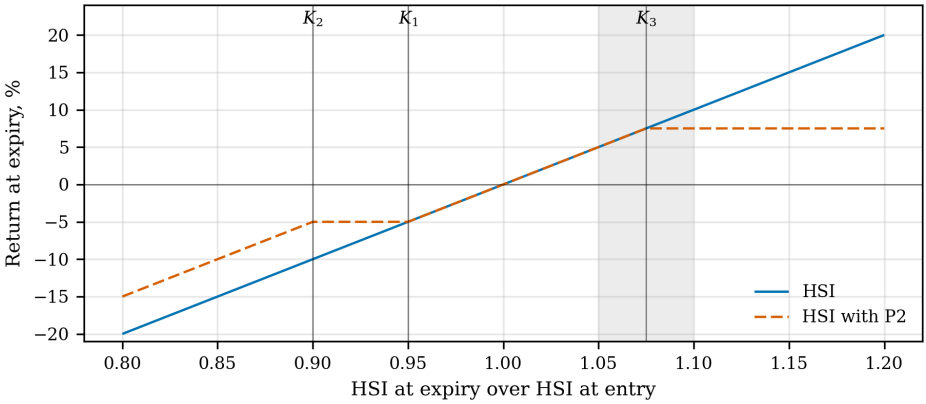
$$G \approx 25\, n \sum_t \Gamma_t S_t^2 \left(\frac{(v_0/100)^2}{A} - r^2_{\mathrm{HSI},t+1} \right).$$

- P1 sells n out-of-the-money calls at the first listed strike at or above $1.04S$ and hedges their delta with HSI futures reset at every close.
- Γ_t is the gamma of one call, v_0 the VHSI at which the calls are sold and A the number of trading days per year.
- So P1 gains when the index moves less than the VHSI implied.

Derivatives strategies

- The protection strategy P2 limits losses in falling markets at a low cost. It holds a put spread collar only when the VHSI signals stress.
- P2 triggers when the VHSI closes above 30 and exits when it closes below 22.
- Each position buys n puts at the first listed strike K_1 at or below $0.95S$ and sells n puts at the first listed strike K_2 at or below $0.90S$.
- It also sells n calls at the first listed strike K_3 at or above the strike within $[1.05S, 1.10S]$ that sets the net premium to zero.
- The size is capped at entry so that the absolute delta does not exceed 15% of the net asset value.

Return at expiry with the P2 collar



Return at expiry of the HSI and of the HSI with the P2 collar.

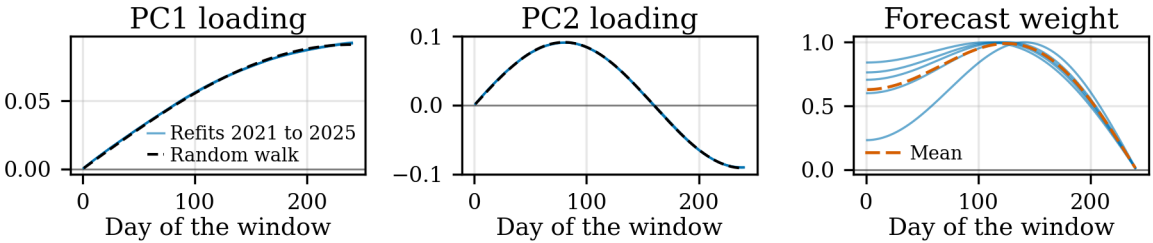
Rank IC of each forecast

	Mean IC by year					All months		
Embedding	2021/22	2022/23	2023/24	2024/25	2025/26	Mean	t	IC > 0, %
PCA	0.094	0.054	0.136	0.035	0.040	0.072	2.47	63
VAE ($d = 2$)	0.098	0.069	0.138	0.039	0.047	0.078	2.66	63
VAE ($d = 3$)	0.084	0.050	0.026	0.024	0.040	0.045	1.49	58

Rank IC of each forecast, averaged within each year from September to August and over all 60 months, with the t-statistic of the mean and the share of months with a positive IC.

- Both embeddings carry predictive information about next-month returns, with mean ICs of 0.072 ($t = 2.47$) for PCA and 0.078 ($t = 2.66$) for the VAE with $d = 2$.
- Their monthly ICs have a correlation of 0.99, although the IC of the VAE is marginally higher in every year.
- The VAE with $d = 3$ is weaker and not significant.

Loadings and forecast weights



Loadings of the first two principal components at each refit with the eigenvectors of a random walk, and the forecast weight η_g relative to the largest weight.

- At every refit, the first two loadings coincide with the eigenvectors of a random walk, and their shares of the variance, about 80% and 9%, match its 81% and 9%.
- The forecast sums standardized daily returns with weights η_g (Appendix F.1), largest for returns from 3 to 9 months before the forecast and small for the most recent month. So the forecast reflects momentum over the past year (Jegadeesh and Titman, 1993).

Performance of the stock portfolios

Appendix D defines annualized return, volatility, Sharpe ratio, maximum drawdown (MDD) and, against a benchmark, active return, tracking error (TE) and information ratio (IR).

Portfolio	Return	Volatility	Sharpe ratio	Max drawdown	Turnover	Avg. # holdings
EW	2.0	24.1	−0.04	−39.8	8.0	78
M1	2.2	25.3	−0.03	−43.0	18.6	42
M2-PCA	5.3	23.9	0.09	−40.6	20.4	40
M2-VAE	4.7	24.0	0.07	−40.6	21.8	40
M3	3.8	24.0	0.03	−40.7	25.6	40
HSI TR	3.1	24.6	0.00	−41.7		

Performance of the stock portfolios from September 2021 to August 2026.

- The backtest shows that stock selection turns the forecast into outperformance, whereas portfolio construction alone does not.
- EW and M1 underperform the HSI TR, while M2-PCA and M2-VAE outperform it with lower volatility.

Active performance of the stock portfolios

	Against the HSI TR					Against M1		
Portfolio	Beta	Active	TE	IR	t	Active	IR	t
EW	0.94	−1.0	6.7	−0.16	−0.35			
M1	1.01	−0.8	5.2	−0.16	−0.36			
M2-PCA	0.95	+2.1	5.2	+0.40	+0.89	+2.9	+0.76	+1.70
M2-VAE	0.95	+1.5	5.2	+0.29	+0.65	+2.4	+0.62	+1.40
M3	0.95	+0.7	5.2	+0.14	+0.30	+1.6	+0.39	+0.88

Active performance of the stock portfolios against the HSI TR and against M1.

- Against M1, which isolates stock selection, M2-PCA and M2-VAE earn active returns of 2.9% ($t = 1.70$) and 2.4% ($t = 1.40$).
- The difference between M2-PCA and M2-VAE, 0.57% a year ($t = 1.24$), is not significant.
- The M2 portfolios maintain a beta of 0.95, so the gain is not a market bet.

Derivatives strategies on M2-PCA

		Performance				Over none		Over HSI TR		
Portfolio	Scenario	Return	Vol.	Sharpe	MDD	Gain	t	Active	IR	Days
M2-PCA	None	5.24	22.8	0.10	−39.2			+2.06	0.38	
	P1	5.97	22.9	0.13	−38.6	+0.73	1.43	+2.75	0.51	
	P2	5.55	21.1	0.12	−35.5	+0.31	0.21	+2.36	0.36	31
	P1 and P2	6.28	21.2	0.15	−34.9	+1.04	0.68	+3.05	0.46	31
	P1 unhedged	5.02	19.9	0.10	−34.3	−0.22	−0.10	+1.85	0.23	
	P2, full trigger	3.24	19.7	0.01	−35.8	−2.00	−1.04	+0.15	0.02	85

Derivatives strategies on M2-PCA at coverage $h = 0.5$. Gain is the difference in annual return from no derivatives, active return and IR are measured against the HSI TR, and Days is the share of days on which P2 is held, in %. P1 unhedged holds the calls of P1 without futures, and P2 on the full trigger adds the drawdown condition of the investment proposal.

- P1 raises returns by 0.73% a year at unchanged volatility ($t = 1.43$), whereas the covered call (P1 unhedged) reduces returns.
- P2 raises the return by 0.31% a year and lowers the MDD from 39.2% to 35.5%, whereas the full trigger lowers the return by 2.00% a year.

Sources of the active return

		M2-PCA			M2-VAE		
Source	Comparison	Active	IR	t	Active	IR	t
Portfolio construction	M1 against the HSI TR	−0.85	−0.16	−0.36	−0.85	−0.16	−0.36
Stock selection	M2 against M1	+2.93	+0.76	+1.70	+2.35	+0.62	+1.40
Derivatives, P1	P1 against none	+0.69	+0.64	+1.43	+0.69	+0.64	+1.42
Derivatives, P2	P1 and P2 against P1	+0.29	+0.09	+0.21	+0.30	+0.10	+0.21
Total	P1 and P2 against the HSI TR	+3.05	+0.46	+1.03	+2.50	+0.38	+0.86

Sources of the active return of M2-PCA and M2-VAE with P1 and P2 against the HSI TR at $h = 0.5$. Each step is measured against the previous one, and the steps add up to the total except for the cash held for margin, which changes the active return by −0.02% for M2-PCA and +0.01% for M2-VAE. None is the portfolio without derivatives.

- For M2-PCA, portfolio construction alone loses 0.85% a year against the HSI TR, stock selection adds 2.93%, the main source of the outperformance, and derivatives add 0.98%.
- Although the forecast itself is significant, its transfer into active return is not, and neither are the derivatives strategies.

Conclusion

- We embed each stock's trading history with PCA and an LSTM VAE. The embeddings carry significant information, and their forecast reflects momentum over the past year.
- The VAE and PCA perform similarly, because the standardized return path behaves like a random walk that leaves little structure beyond its principal components.
- From September 2021 to August 2026, the PCA portfolio with derivatives strategies outperforms the HSI TR by 3.05% a year, mainly from stock selection.
- However, neither the transfer of the forecast into active return nor the gain from the derivatives is statistically significant.

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Code and replication data are available at *GitHub*.

Appendix A.1. Diagnostics of the VAE embeddings

		Active units		R^2 on PCA scores	
Refit	Training windows	$d = 2$	$d = 3$	Dimension 1	Dimension 2
2021	154,401	2	2	0.91	0.97
2022	167,233	2	2	0.91	0.97
2023	182,816	2	3	0.94	0.97
2024	201,166	2	2	0.68	0.91
2025	220,792	2	2	0.92	0.96

Diagnostics of the VAE embeddings at each refit. Active units count the latent dimensions whose posterior mean has a variance across windows above 0.01 (Burda et al., 2016). R^2 is that of each dimension of the embedding of the VAE with $d = 2$ regressed on the two PCA scores of the same windows.

- The PCA scores explain the VAE embedding with $R^2 > 0.9$ in four of the five refits.
- With $d = 3$, one dimension is inactive in four refits, which adds nothing and lowers the IC.

Appendix D.1. Performance measures

$$\text{Return} = e^{A\bar{r}^{\text{p}}} - 1, \qquad \text{Vol} = \sqrt{A}\, s(r^{\text{p}}),$$

$$\text{SR} = \sqrt{A}\,\frac{\bar{e}^{\text{p}}}{s(e^{\text{p}})}, \qquad \text{MDD} = \min_t \left(\frac{W_t}{\max\left(1, \max_{s \le t} W_s\right)} - 1 \right),$$

- r_t^{p} and r_t^{b} are the daily returns of a portfolio and of a benchmark over Y years with A trading days per calendar year.
- $e_t^{\text{p}} = r_t^{\text{p}} - r_t^f$ and $e_t^{\text{b}} = r_t^{\text{b}} - r_t^f$ are their excess returns and $W_t = \exp(\sum_{s \le t} r_s^{\text{p}})$.
- $s(x)$ and $\bar{x}$ are the sample standard deviation and mean of a daily series x_t.
- Return, Vol and SR are the annualized return, volatility and Sharpe ratio.

Appendix D.2. Performance measures against a benchmark

$$\text{Active} = A\,\bar{\tilde{r}}, \qquad \text{TE} = \sqrt{A}\,s(\tilde{r}),$$

$$\text{IR} = \frac{\text{Active}}{\text{TE}}, \qquad t_{\text{IR}} = \text{IR}\sqrt{Y},$$

$$\beta^{\text{p}} = \frac{\sum_t \left(e_t^{\text{p}} - \bar{e}^{\text{p}}\right)\left(e_t^{\text{b}} - \bar{e}^{\text{b}}\right)}{\sum_t \left(e_t^{\text{b}} - \bar{e}^{\text{b}}\right)^2},$$

- $\tilde{r}_t = r_t^{\text{p}} - r_t^{\text{b}}$, and the other symbols are as in Appendix D.1.
- t_{IR} is the t-statistic of the active return and β^{p} the beta.
- A two-sided test at 5% requires $|t_{\text{IR}}| \geq 1.96$, that is $\text{IR} \geq 1.96/\sqrt{Y} = 0.88$ for the $Y = 4.99$ years of the backtest.

Appendix E.1. Derivatives strategies at full coverage

		Performance				Over none		Over HSI TR		
Portfolio	Scenario	Return	Vol.	Sharpe	MDD	Gain	t	Active	IR	Days
M2-PCA	None	5.21	21.8	0.10	−37.7			+2.04	0.36	
	P1	6.60	22.0	0.16	−36.6	+1.38	1.42	+3.35	0.57	
	P2	5.55	20.0	0.12	−33.9	+0.34	0.23	+2.36	0.33	31
	P1 and P2	6.94	20.2	0.19	−32.7	+1.72	0.99	+3.66	0.50	31
M2-VAE	None	4.66	21.9	0.07	−37.7			+1.51	0.27	
	P1	6.05	22.1	0.13	−36.5	+1.39	1.42	+2.83	0.49	
	P2	5.00	20.1	0.10	−33.9	+0.34	0.23	+1.84	0.26	31
	P1 and P2	6.39	20.3	0.16	−32.6	+1.73	0.99	+3.15	0.44	31

Derivatives strategies on M2-PCA and M2-VAE at coverage $h = 1$, with columns as at $h = 0.5$.

- On M2-PCA, P1 adds 1.38% a year, P2 adds 0.34% and both together add 1.72%, for an active return of 3.66% against the HSI TR with an IR of 0.50.
- The contribution of P1 scales with its size, while the delta cap binds at all 19 entries of P2, which therefore changes little.

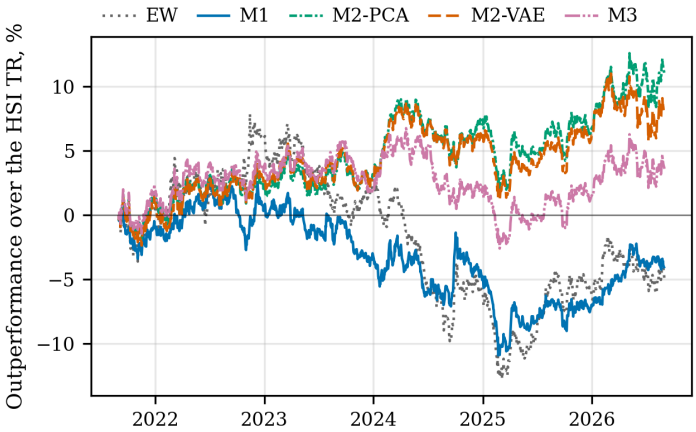
Appendix F.1. Forecast weights

Because the path is cumulative, the forecast is a weighted sum of standardized daily returns,

$$\hat{u}_{i,k} = \text{const} + \sum_{g=1}^{L} \eta_g \, \frac{r_{i,\tau_k-1-L+g}}{s_{i,\tau_k-1}}, \qquad \eta_g = \frac{1}{\sqrt{L}} \sum_{j=g}^{L} \left(\hat{b}_1 \chi_{1,j} + \hat{b}_2 \chi_{2,j} \right).$$

- $\chi_{1,j}$ and $\chi_{2,j}$ are the loadings of the two components and $\hat{b}_1$ and $\hat{b}_2$ their coefficients.
- The first component weights the whole window and tracks the volatility-scaled return over the year.
- The second contrasts the first two-thirds of the window with the last third and so measures whether the gains came early or late.

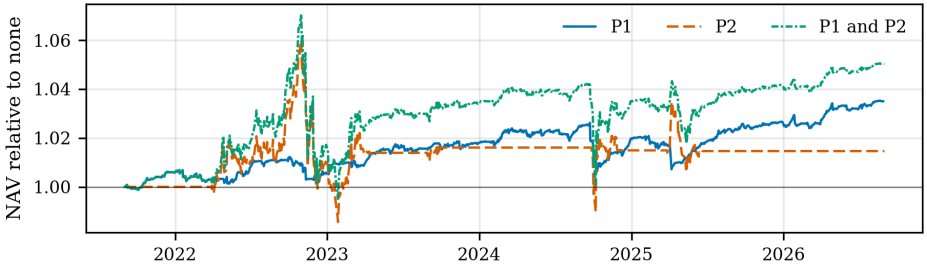
Appendix G.1. Outperformance over the HSI TR



Outperformance of each stock portfolio over the HSI TR, in %.

- The outperformance of M2-PCA and M2-VAE increases over time.
- It does not arise in any specific market environment.

Appendix G.2. Net asset value with derivatives strategies



Net asset value of M2-PCA with derivatives strategies relative to that without derivatives.

- The gain of P2 arises in the decline of 2022.
- The collar was held from March 2022 to April 2023, and its short call gave up the whole gain in the rebound from November 2022 before the collar closed ahead in April 2023.